

MSc Research Project/Internship/ Practicum

Handbook Document

Handbook Overview

An important resource for Research Project/Internship/ Practicum module

Available on *MSc Research Project/Internship/ Practicum Info & Submission* Moodle page

Provides guidelines to students regarding

- Role of supervisor
- How the student should manage the Research Project/Internship/Practicum project
- What is required to be submitted by the student
- Structure of the research paper style report
- Other aspects: request for deferral, plagiarism, useful contacts
- Various Forms: Ethics related Forms, Exist Award, Internship Contract Tri-party, etc.

Supervisory Relationship

- **Supervisor provides adequate guidance to a student on the production of their Research Project/Internship/Practicum**
- **Always have some written material to bring along**
- **Student/Supervisor Expectations (more details in the Handbook document)**
- **Attend all supervision classes as scheduled in the Timetable**

Research Paper Style Report

- A research paper style report describes the individual research project specification and software/artefact development as well as the testing/evaluation results.
- A literature review should also be included to ensure the validity of the research area. Students are also required to critically analyse insights gained throughout evaluation of their software/artefact development.
- **The work presented in this report shall build upon the work conducted as part of Research in Computing module/ Practicum Part 1 (except for MSc students doing Internship Part 2)**

Research Paper Style Report

- **Report structure: The following sections must exist**
 - Abstract
 - Introduction
 - Literature Survey
 - Research Methodology
 - Design and Implementation Specifications
 - Evaluation
 - Conclusions and Discussion
 - References
- **Student **MUST** use the provided **report template** (MSc Word or LaTeX) available on Moodle page**

Report Structure

Abstract

- The role of the abstract is to summarize the project you have undertaken in a concise manner, highlighting the motivation for the project, the objectives, the methodology, the results and findings.

Report Structure

Introduction

- This section should set the scene for the reader.
- The background to the problem should be described, with the motivations for the choice of project, what gap in the literature it seeks to fill, and identification of the project's beneficiaries (academic or industrial).
- Your research question(s) and objectives need to be precisely stated, together with the tests that will show, at the end of the project, that they have been met (or not been met).
- You need also to outline your methods in broad terms to show how you planned to meet the objectives.
- Finally, outline the structure of the report, showing how it fits together.
- Length: Up to 2 pages

Report Structure

Review of the Literature/Background/ Literature Survey

- The updated literature review should demonstrate evidence of independent research, critically analyse the potential of an application / idea and provide insights into how it can be implemented and evaluated.
- This component shall build upon the literature review work conducted as part of Research in Computing module and shall be updated and revised based on feedback from supervisors and based on the final developed product.
- Length: 3-4 pages

Direct copy-paste or paraphrasing from the Research in Computing (RiC) / Practicum Part 1 proposal is not permitted as you already received marks and credits for that work.

Failure to update and expand substantially the sections that overlap with the RiC / Practicum Part 1 proposal will result in penalties and mark deductions.

Report Structure

Outputs Summary:

- For each output produced, please provide a brief overview of the output, the output type (e.g. code, design artefacts, models, test cases) and potential user(s) of the output.
- It should be a maximum of 2 pages.

Report Structure

Methodology

- This section describe in detail the methods for whatever activities were necessary for your project – e.g., requirements/contextual analysis, data gathering, data cleaning, data analysis, implementation, testing/evaluation, etc.
- Your choice of methods should be discussed and justified with respect to the research question and objectives.
- Report not only what methods you applied in generic terms, but sufficient information details for the reader to understand how you ran your project.
- Report what you did, not what you produced or found as a result (which goes under Results).

Report Structure

Design Specification:

- The techniques and/or architecture / framework that underlie the implementation and the associated requirements are identified.

Implementation/Solution Development Specification:

- The purpose is to describe only the final stage of the implementation.
- It should describe the outputs produced, e.g. transformed data, code written, models developed, questionnaires administered.
- The description should also include what tools and languages you used to produce the outputs.
- It must not contain extensive code listing or a user manual.

Report Structure

Results and Critical Analysis:

- A comprehensive analysis of the results and main findings of the study as well as the implications of these finding both from academic and practitioner perspective are presented.
- For example:
 - an algorithm may be comprehensively evaluated using novel performance specific metrics on the Cloud Competency Centre facilities or external cloud providers;
 - a data mining approach could be evaluated using standard techniques such as accuracy, sensitivity and error rate,
 - an internet application or mobile application may be evaluated using suitable usability testing techniques.
- Statistical tools should be used to critically evaluate and assess the experimental research outputs and levels of significance.

Report Structure

Results and Critical Analysis:

- It is imperative that you critically analyse your results and discuss how they answer your original research question.
- You must also compare your results with previous work that you have reviewed in the literature.
- It is usually very helpful for the readers to include tables, graphics and diagrams to identify key trends and relationships results. If you do so, be sure to refer to these figures in the text and use them as evidence to support what you are explaining or arguing.
- Due to page number limitation only the most relevant results are analysed and presented as an evidence to support or reject the hypothesis.

Report Structure

Discussion and conclusions:

- This section should examine your results in comparison with your objectives. Discuss your confidence in the results, their validity, scope and their generalisability. What are the implications of what you have found out?
- Discuss any proposals that you might make for further work, having discovered what you now know.

Report Structure

References:

- Correct citation references must be included
- Contains a full list of any source directly referred to or quoted in the text following the format provided in the research project template.
- Use tools to assist you with both in-text referencing and the presentation of their reference lists. Please consult these tools available from NCI library .
- As the research project report should not have more than 20 pages you must select and refer in the text the most relevant references that related to your research topic and work.

Configuration manual:

- Should be provided as a separate document to the research paper style report and it will not count towards the suggested report page limit

Report Structure

Other Considerations:

- Remember that all figures, tables, photographs etc. should be referenced and numbered in the report e.g. the graph (Fig 1) which complements the Table (Table 1).
- It is highly recommended that students take some time at the start of the research process to go to the library and review other MSc research projects. This will give you a feel for the type of appropriate research questions, the research approaches adopted and the format of the report.